Internal Q&A — Layer 2

UIO–NN paper | concise defense notes

Core position. Layer 1 has the main theoretical guarantee. Layer 2 has a *conditional* $\mathcal{L}_2$ observer-error certificate. The paper does **not** prove convergence or boundedness of online NN weights θ, reduction of Δ_x, or full closed-loop stability.

Q. Do you prove stability of Layer 2?

Say: We prove a conditional $\mathcal{L}_2$ error bound for the Layer-2 observer. With $e = x - \hat{x}_{NN}$ and $\Delta_x = \mu_x - \mu_\theta(\hat{x}_{NN})$, LMI feasibility gives

$$\dot{V} + \|e\|^2 - \sigma\|\Delta_x\|^2 < 0, \qquad V = e^\top P e.$$

Thus $\|e\|_{\mathcal{L}_2} \leq \sqrt{\sigma}\|\Delta_x\|_{\mathcal{L}_2} + \sqrt{\lambda_{\max}(P)}\|e(0)\|$.

Limit: This is **not** a proof of NN-learning stability: it does not show θ converges, remains bounded, or makes $\Delta_x \to 0$.

Q. Is the Layer-2 error exponentially stable?

Say: Only in the ideal residual-free case. If $\Delta_x = 0$, the strict Lyapunov inequality gives $\dot{V} \leq -\|e\|^2 \leq -V/\lambda_{\max}(P)$, hence exponential convergence.

Limit: For nonzero Δ_x, the stated result is an $\mathcal{L}_2$ gain bound. Do not call it unconditional exponential stability or a pointwise state bound.

Q. How is the LMI proof obtained?

Say: Write $\dot{e} = (\nabla\varphi_x - L_{NN}C)e + B\Delta_x$, use $V = e^\top P e$, set $L_{NN} = P^{-1}Q^\top$, and require the resulting quadratic form to be negative at every Jacobian-polytope vertex. The Schur complement gives the LMI; convexity covers all admissible Jacobians; integration gives the $\mathcal{L}_2$ bound.

Q. What does the NN contribute to that theorem?

Say: The theorem certifies robustness of the observer gain against the residual Δ_x. The NN can improve performance only by reducing that residual, which is evaluated in simulation.

Limit: The LMI by itself does not establish that the NN is beneficial; the same certificate applies to any additive residual input.

Q. Why use a UIO estimate as NN supervision?

Say: When the true tire-force residual is unavailable, Layer 1 supplies a bounded physics-based proxy target. Layer 2 uses that signal together with measurements to refine the estimate.

Limit: The proxy can be biased, especially when the rank condition is relaxed. It is supervision, not ground truth.

Q. Does the NN have a shortcut because $\hat{\mu}_x$ is both an input and its target?

Say: Yes, this is a legitimate concern. As written, $\mathcal{N}_\theta(\hat{x}_{NN}, \hat{x}_{UIO}, \hat{\mu}_x, u)$ can learn to copy $\hat{\mu}_x$. If retained, describe Layer 2 as a **learned UIO-refinement/filter**, not necessarily a predictive state-dependent dynamics model.

Limit: To support a predictive-dynamics claim, train/evaluate a version without the current $\hat{\mu}_x$ input (for example using state and input, or past UIO history only) and report the ablation.

Q. Why is x_{ref} included in the training loss? Does it bias the estimate?

Say: It acts as a task-specific prior that encourages useful estimates for trajectory tracking.

Limit: It can bias the observer toward the desired trajectory rather than the actual disturbed state. Do not present it as independent evidence of estimation accuracy; report state-estimation errors and an ablation without this term.

Q. Do you prove closed-loop vehicle–controller stability?

Say: No. We certify an observer-error property. Closed-loop stability of the plant, controller, NN adaptation, and both observers would require a separate ISS/small-gain or composite-Lyapunov analysis.

Q. What is the role of the rank relaxation and output derivative?

Say: If $\text{rank}(CB) = \text{rank}(B)$, an exponential UIO can be constructed with the output-derivative augmentation. When it fails, regularization trades exact unknown-input decoupling for an adjustable $\mathcal{H}^1$ error bound.

Q. What evidence would make the Layer-2 claim convincing?

Say: Show ablations on the same manoeuvres and unseen manoeuvres: (i) UIO only, (ii) UIO plus a simple low-pass filter, (iii) NN without UIO supervision, and (iv) the full method. Include sensor noise, model mismatch, and the NN-input shortcut test.

One-sentence presentation claim. "Layer 1 provides a bounded physics-based supervisory estimate; Layer 2 is a practical residual-refinement observer with a certified $\mathcal{L}_2$ error gain conditional on the remaining approximation residual."